EXP-PFDR-2d36f9 break-even growth ladder

Run report — 2026-10-31

Experiment	EXP-PFDR-2d36f9
Hypothesis	H-PFDR-ed7ccf
Goal	GOAL-PFDR-fd09e0
Run IDs	RUN-PFDR-f86a2a, b507a1, 601709, b126f3, 97fa6c, 8cb633, 73276d
Code commit	41020cfad0 (clean)
Reading	**O-INVALID**

0.1. What ran

Seven planned trials, all with `output_validated` execution receipts: selftest, the descriptive reanalysis of EXP-PFDR-ca5e01, pilot, census-main (14,571 jobs), analyze-main, census-rep (no ALIVE lines, so it records an empty result and stops), and analyze-final. The selftest passed 126 kernel-vs-brute checks, 9 near-2^32 key checks, 6 `plant_internal` checks and 12 order certifications including 8 at 32 bits. Census-main recorded 0 failed-infrastructure jobs.

0.2. Recorded structured-line values

The five-rung slope `delta_hat` and the frozen class are recorded below. Three of the six lines are invalid because the pilot's mean null count at bits 16 did not reach the frozen 2500-coincidence target; no class is assigned to them.

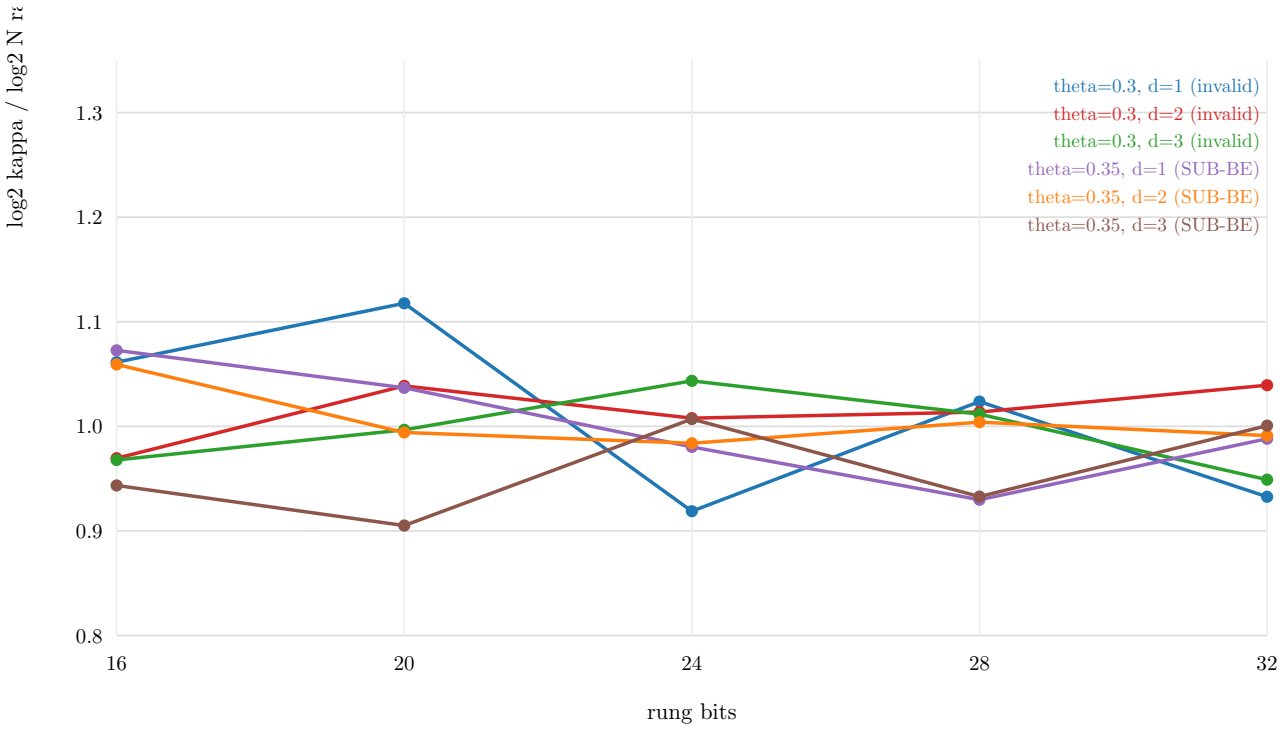

Figure 1: Recorded kappa per rung for the six structured lines (RUN-PFDR-b126f3, RUN-PFDR-73276d). Points are the measured mean kappa at each rung; lines connect recorded values only. Classes are the frozen analysis labels.

theta	d	class	delta_hat	reason
0.3	1	invalid	−0.012502	b16-t0.30-d1 M 2333.0 < 2500
0.3	2	invalid	0.004153	b16-t0.30-d2 M 2333.0 < 2500
0.3	3	invalid	−0.000893	b16-t0.30-d3 M 2333.0 < 2500
0.35	1	SUB-BE	−0.009795	valid
0.35	2	SUB-BE	−0.004440	valid
0.35	3	SUB-BE	0.005368	valid

0.3. Recorded controls

control	pass	entries	design and slope ok
C3	False	8	4
C4	False	2	0
C5	False	6	0
C6	False	0	0

Calibration recorded n = 30, mean = −0.149520, sd = 0.659750, pass = False. The analysis therefore records `O-INVALID`: fewer than five of six structured lines are valid and the blocking calibration/controls do not pass. This report describes the recorded run; it does not decide support, weakening, or the fate of H-PFDR-ed7ccf.

0.4. Missing and derived values

The raw census job file is stored in the artifact store as `main-jobs.jsonl.gz` with a `.artifacts.json` manifest; the report graphs only values recorded in `analysis.json`. `delta_hat` is the frozen ordinary-least-squares slope of log2 kappa on the rung mean log2 N; the bootstrap interval is recorded in `analysis.json`, and is not recomputed here.